

Part 1 | Project Proposal

Predicting student dropout and academic success

CS-13410 Introduction to Machine Learning · Fall 2026 · Assignment 1

Problem and target

Predict a university student's eventual **Dropout, Enrolled, or Graduate** outcome using data known by the end of the second semester. This is a three-class classification problem. The target is the CSV column **Target**. The semester outcomes are available at the proposed prediction time; this is not an enrollment-time forecast.

Dataset and suitability

The supplied data.csv matches the UCI “Predict Students' Dropout and Academic Success” dataset: 4,424 students, 36 predictors, and one target. It combines academic counts/grades, demographics, application categories, and economic indicators. Its size and mixed numeric/categorical features support the required preprocessing and later model comparisons. Classes: Graduate 2,209; Dropout 1,421; Enrolled 794. There are no missing cells or exact duplicate rows in this supplied copy. UCI lists the dataset under CC BY 4.0.

Success measure

Use **macro F1** as the primary metric so all three outcomes receive equal weight. Later model evaluations will also report per-class recall, especially for the smaller Enrolled class. A stratified 80/20 train/test split (random state 42) reserves the test partition for final evaluation; cross-validation in later parts will occur on training data.

Part 1 preparation plan

Create first- and second-semester approval rates. Median-impute continuous features, cap numeric outliers using train-fitted IQR bounds, and robust-scale them. Most-frequent-impute categorical fields and one-hot encode them; preserve binary flags. All operations are inside a scikit-learn Pipeline/ColumnTransformer fitted to training data only. No global resampling is done. Part 2 will address imbalance with stratified folds and model-level weighting where appropriate.

Limitations and next steps

Semester-2 records can only support prediction after that semester. Academic measurements may strongly reflect outcomes already taking shape, so deployment timing must be explicit. The data originate in Portuguese higher education; performance and fairness may not transfer to other institutions. The team reports that the instructor has approved the dataset. Group registration is a separate course requirement. Parts 2–4 will extend the same notebook with models and evaluation.

Source: Realinho et al. (2021), UCI Machine Learning Repository, doi:10.24432/C5MC89. License: CC BY 4.0.